

German University in Cairo

Faculty of Engineering and Materials Science

NETW1010 — Internet of Things

Milestone 1 Report

BLE-Based Physiotherapy Tracking and Assessment

*Use Case: Elbow Flexion / Extension
Rehabilitation*

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1. Project Description

1.1 What Problem Are We Solving?

When a patient undergoes elbow surgery or suffers an elbow injury, they are typically prescribed a set of physiotherapy exercises to be performed daily at home. These exercises often involve controlled flexion and extension of the elbow joint to restore mobility and strength.

However, a major challenge arises because there is **no reliable way to monitor whether the patient is performing these exercises correctly or consistently** outside the clinic. The physiotherapist only evaluates the patient during periodic visits, leaving all home sessions unobserved.

This lack of monitoring leads to several issues:

- Patients may perform exercises with **incorrect range of motion**
- They may **not reach the required target angle**
- They may **skip sessions or underperform repetitions**

For example, if a patient only bends their elbow to 60° instead of the required 100°, their recovery progress will be significantly affected — yet neither the patient nor the physiotherapist is aware of this in real time.

1.2 Who Are the Users?

- **Patients:**
Individuals recovering from elbow injuries or surgeries who are required to perform daily rehabilitation exercises at home. They will wear the proposed wearable device on their forearm.
 - **Physiotherapists:**
Medical professionals responsible for prescribing and monitoring rehabilitation exercises. They will access patient data remotely through a web-based dashboard.
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1.3 What Is Our Solution?

We propose a **BLE-based wearable wristband** built using an **nRF52 (Saeed) microcontroller**, which includes an **integrated IMU (Inertial Measurement Unit)** consisting of an accelerometer and gyroscope.

Unlike traditional designs that rely on external sensors such as MPU-6050, our system utilizes the **onboard IMU sensor** available on the nRF board. This reduces hardware complexity, improves reliability, and makes the device more compact and power-efficient.

The device is worn on the patient's forearm and performs the following functions:

- Measures arm movement using the onboard IMU
- Calculates the **elbow joint angle in real time**
- Counts repetitions automatically
- Displays live feedback (angle and reps) on a small screen
- Transmits data via **Bluetooth Low Energy (BLE)** to a nearby gateway (phone or laptop)

The gateway then forwards the data to a cloud-based system, where the physiotherapist can monitor patient performance through a web dashboard.

The interaction is kept extremely simple for the patient:

- Press button → start session
 - Perform exercises
 - Press button → end session
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1.4 Why Is This Useful?

- Provides **objective, real-time data** instead of relying on patient self-reporting
 - Enables **remote monitoring** by physiotherapists
 - Gives **instant feedback** to patients during exercise
 - Improves **exercise accuracy and recovery outcomes**
 - Uses **low-power BLE**, making it ideal for wearable devices
 - Reduces hardware cost and complexity by using the **integrated IMU on the nRF board**
 - Enables detection of incorrect exercise execution and missed sessions
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1.5 What Do We Expect to Achieve?

1. A fully functional wearable wristband based on **nRF52 with onboard IMU** capable of measuring elbow angle
2. Accurate **real-time angle calculation** using accelerometer and gyroscope data
3. Automatic **repetition counting** based on motion thresholds
4. Reliable **BLE communication** with a phone or laptop
5. A cloud-based dashboard for **visualizing patient progress**
6. A system capable of generating **alerts for incorrect exercise or inactivity**

2. System Architecture

The proposed system consists of four main components that operate together to provide continuous monitoring of physiotherapy exercises. Data flows from the wearable wristband to the physiotherapist through a structured pipeline, enabling real-time tracking and long-term analysis.

2.1 The Wristband (nRF Microcontroller + Onboard IMU)

This is the wearable device placed on the patient's forearm. It is built around an **nRF52 (Saeed) microcontroller**, which integrates **Bluetooth Low Energy (BLE)** communication and an onboard **Inertial Measurement Unit (IMU)**.

The onboard IMU (e.g., LSM6DS3 IMU sensor) combines:

- A 3-axis accelerometer
- A 3-axis gyroscope

This eliminates the need for an external sensor such as MPU-6050, reducing hardware complexity and improving system reliability.



Main Functions:

- The nRF microcontroller samples IMU data at approximately **50 Hz**
- Processes raw acceleration and angular velocity data
- Computes the **elbow joint angle in real time**
- Detects and counts completed repetitions
- Displays angle and rep count on a small OLED screen
- Transmits data via BLE to the gateway device



Calibration:

At the start of each session:

- The patient fully extends their arm
- This position is stored as the **0° reference**
- All subsequent movements are measured relative to this baseline



Sensor Placement:

The wristband is worn on the forearm, just below the elbow joint, allowing accurate tracking of elbow flexion and extension.

2.2 The Gateway (Phone or Laptop)

The gateway is the patient's existing **smartphone or laptop**, which acts as an intermediary between the wearable device and the cloud.

Functions:

- Receives data from the wristband via **BLE**
- Parses incoming data (angle, repetitions, session info)
- Forwards the data to the cloud server via Wi-Fi

A lightweight background application (e.g., Python script or mobile app) handles this process automatically without requiring user interaction.

2.3 The Cloud Server

The cloud server is responsible for storing, processing, and managing all incoming physiotherapy data.

Implementation:

- Built using a backend framework such as Node.js
- Connected to a database (e.g., MongoDB or SQL)

Stored Data Includes:

- Session timestamps
- Angle measurements over time
- Number of repetitions
- Maximum achieved angle
- Warnings or incorrect movement flags

This centralized storage allows long-term tracking of patient recovery progress.

2.4 The Dashboard (Web Application)

The dashboard is a web-based interface used by physiotherapists to monitor patients remotely.

Features:

- Secure login system for clinicians
- Patient selection and session history
- Graphical visualization of elbow angle over time
- Display of repetition counts and performance metrics
- Identification of missed sessions or insufficient movement range

This interface enables physiotherapists to make informed decisions and adjust rehabilitation plans without requiring frequent in-person visits.

Overall Data Flow

1. IMU sensor captures motion data
2. nRF processes data and calculates angle
3. Data is transmitted via BLE
4. Gateway receives and uploads data via Wi-Fi
5. Cloud stores and processes data
6. Dashboard displays results to physiotherapist

3. Hardware Components

The table below lists all components required to implement the wearable wristband and the overall system. By using an nRF52-based board with an onboard IMU, the design becomes simpler, more compact, and cost-efficient.

Component	What It Does	How It Connects	Qty
nRF52 Development Board (Saeed)	The main microcontroller of the system. It reads motion data from the onboard IMU, calculates the elbow angle, counts repetitions, and transmits data via BLE.	BLE / Internal Sensor Interface / I2C	1
Onboard IMU Sensor (e.g., LSM6DS3 IMU sensor)	Measures acceleration and angular velocity of the forearm. Used to compute elbow angle and detect movement.	Internal (connected directly to nRF)	1
OLED Screen (0.96")	Displays real-time elbow angle and repetition count to provide immediate feedback to the patient.	I2C	1

Component	What It Does	How It Connects	Qty
LiPo Battery (3.7V)	Provides portable power for the wearable device. One charge supports multiple exercise sessions.	—	1
TP4056 Charger Module	Allows safe charging of the battery via USB.	USB	1
Push Button	Used to start and stop exercise sessions.	GPIO	1
3D-Printed Wristband	Mechanical enclosure that holds all components and attaches to the forearm.	—	1
Laptop or Phone (Gateway)	Receives BLE data from the wristband and uploads it to the cloud server via Wi-Fi.	Bluetooth / Wi-Fi	1

4. Communication Protocols

4.1 Internal Sensor Communication (IMU → nRF)

The onboard IMU (such as the LSM6DS3 IMU sensor) is internally connected to the nRF microcontroller through a digital interface (typically I2C or SPI).

The microcontroller samples sensor data at a fixed rate (approximately 50 Hz), retrieving:

- Acceleration data (Ax, Ay, Az)
- Angular velocity data (Gx, Gy, Gz)

These values are then processed to calculate the elbow angle in real time.

4.2 Bluetooth Low Energy (BLE) — Wristband to Gateway

Bluetooth Low Energy (BLE) is used for communication between the wearable device and the gateway (phone or laptop).

✓ Reasons for Choosing BLE:

- **Very low power consumption** (ideal for wearables)
- Direct connection without requiring Wi-Fi
- Supported by all modern smartphones

Operation:

- The wristband acts as a **BLE peripheral (server)**
- The gateway acts as a **BLE central (client)**
- Data is transmitted using **GATT services**, including:
 - Elbow angle
 - Repetition count
 - Session status

BLE allows continuous real-time streaming of data with minimal battery usage.

4.3 Wi-Fi — Gateway to Cloud

The gateway device (phone or laptop) uses its existing Wi-Fi connection to upload collected data to the cloud server.

This approach avoids using Wi-Fi directly on the wearable device, which would significantly increase power consumption.

4.4 MQTT — Data Transmission to Cloud

MQTT is used as the communication protocol between the gateway and the cloud server.

Advantages:

- Lightweight and efficient
- Suitable for real-time IoT data streaming
- Reduces bandwidth usage compared to HTTP

Operation:

- The gateway **publishes** data (angle, reps, session info)
- The server **subscribes** and processes incoming messages

This ensures fast and reliable delivery of physiotherapy session data.

5. Expected System Workflow

The system operates through the following sequence:

1. Device Initialization

The patient wears the wristband on their forearm and powers it on using the push button. The OLED screen displays initial system status and readiness.

2. Calibration Phase

The patient fully extends their arm and holds the position for approximately 2 seconds. This position is stored as the **0° reference angle**, ensuring accurate measurement relative to the patient's natural posture.

3. Session Start

The patient presses the button to start the session.
The system begins:

- Sampling IMU data
- Calculating elbow angle
- Initializing repetition counter

4. Exercise Execution

As the patient performs flexion and extension:

- The elbow angle is displayed in real time
- Repetitions are automatically counted
- If the target angle is not reached, a warning is shown on the screen

5. Data Transmission

During the session:

- The wristband continuously transmits data via BLE.
 - The gateway receives and prepares data for upload.
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6. Session Termination

The patient presses the button again to end the session.
The wristband displays a summary, including:

- Total repetitions
 - Maximum angle achieved
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7. Cloud Upload

The gateway uploads the session data to the cloud server using Wi-Fi and MQTT.
This process occurs automatically without user intervention.

8. Remote Monitoring

The physiotherapist accesses the dashboard and views:

- Angle vs. time graphs
 - Repetition counts
 - Performance trends
 - Alerts for incorrect execution
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9. Adaptive Feedback

Based on patient performance:

- The physiotherapist can adjust target angles
 - Updated parameters are sent back to the system in future sessions
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